# Deduce the World: How Far Can Reasoning Go?

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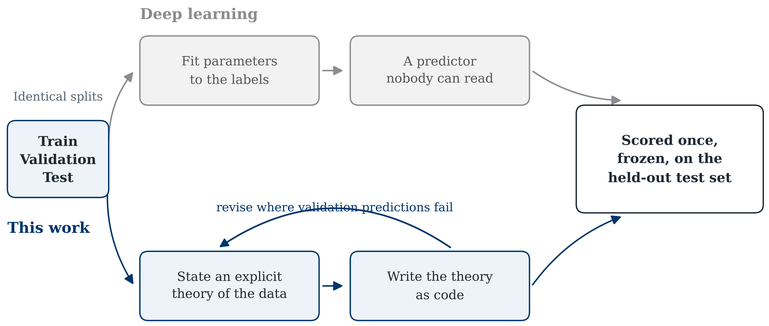
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Results & figures

## 1 Theories, not fitted predictors

Science deduces the world from first principles, revising its theories where their predictions fail. Deep learning does not. It fits a predictor to existing data, which binds it to that distribution.

We ask whether a reasoning model can form scientific theories, and whether those theories generalize better than deep learning.

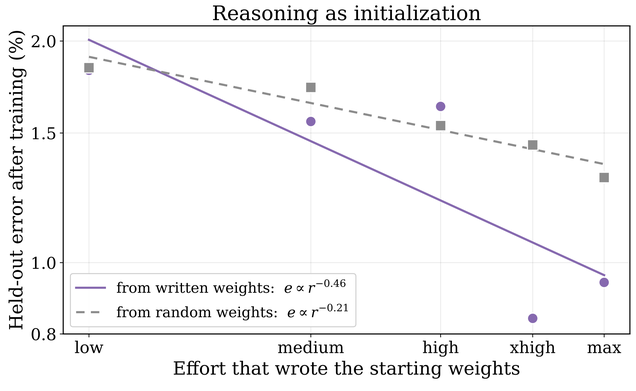



We keep the splits of deep learning and replace the learner. The model states a theory, writes it as code, and revises it against validation.

| | |
|---|---|
| **May** | read papers and the web, measure the training data, inspect its own errors |
| **May not** | fit parameters to labels, search automatically over rules, store examples, see the test split |

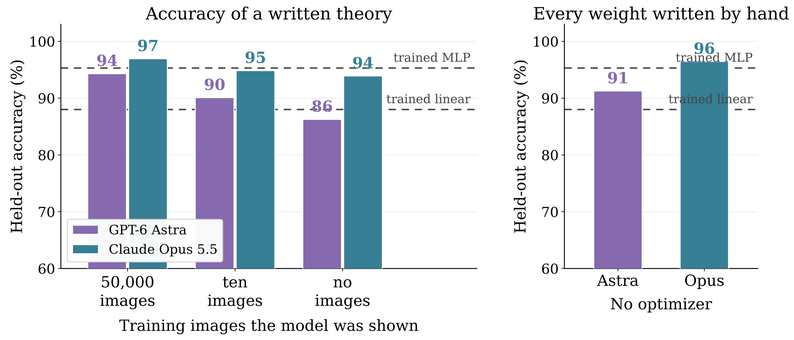
### Reasoning as initialization

A written theory is also a starting point. Training from written weights converges sooner, and by more as reasoning grows.




Held-out error after a fixed training budget against effort rank, both axes logarithmic.

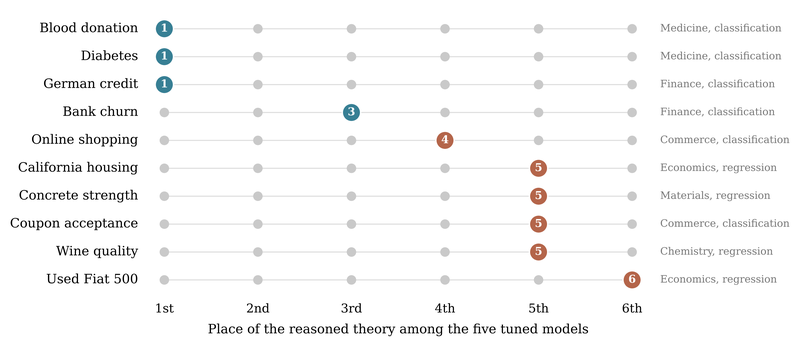
## 2 How much data a theory needs




Handwritten digits. Best of five reasoning efforts. Dashed lines are trained models on the same split.

> Given no training images at all, a theory of how digits are written classifies most of them.

## 3 The same protocol, ten fields




Each task ranks the reasoned theory against linear models, random forests, XGBoost, CatBoost and an MLP, all tuned on the same rows.

> Reasoned theories win on small clinical and credit tables, and trail boosted trees where the signal lies in feature interactions.

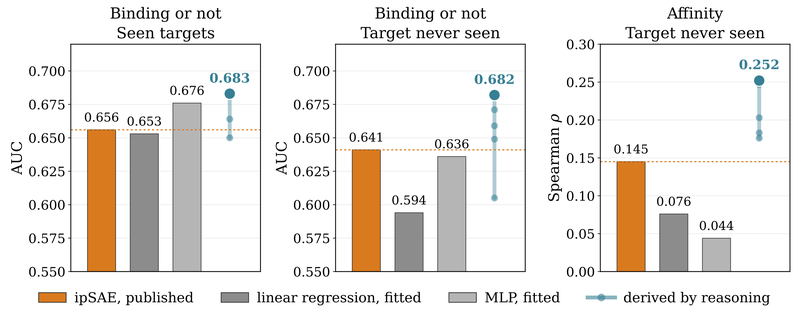
### What a theory looks like

Selected rules from one program on German credit. The points are judgments stated before measurement, not fitted odds ratios.

| | |
|---|---|
| Checking account | none $+3$, at least 200 DM $+1$, negative $-2$ |
| Credit history | critical account elsewhere $+2$, past delay $-1$ |
| Loan duration | $-1$ for each twelve months beyond the first twelve |
| Employment | under a year $-1$, four years or more $+1$ |
| Output | $P(\text{good}) = 1/(1+e^{-(\text{score}-1)/2})$ |

## 4 Deriving a scientific score

A structure predictor returns a complex and an estimate of its own error. Each session receives that matrix and nothing else.



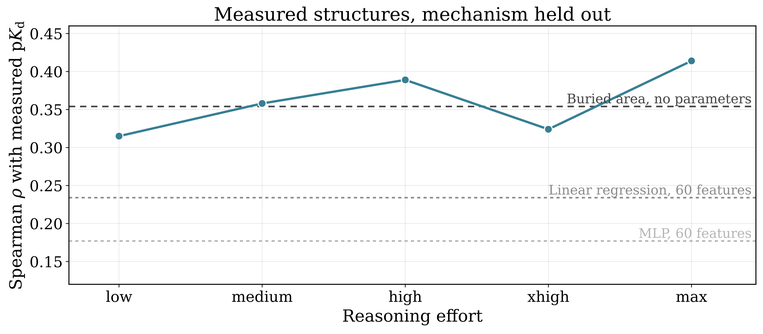

704 designed binders against five targets. The middle and right panels hold out an entire target.

### Derived, not fitted

Treating the bound complex as a harmonic well, with $\sigma_{\text{iface}}$ the precision of the interface and $\sigma_0$ the precision the binder achieves within itself,

$$\kappa = \left(\frac{\sigma_0}{\sigma_{\text{iface}}}\right)^2, \qquad \mathrm{p}K_{\mathrm{d}} = \mathrm{p}K_{\mathrm{d,ref}} + 3\log_{10}\kappa.$$

The 3 is half the six rigid-body degrees of freedom that association confines. Nothing is fitted.




Measured structures grouped by mechanism of association. One mechanism is held out whole.

> What transfers is a dimensionless ratio the pair supplies itself. What fails is anything calibrated on the cases it was fitted to.

Caveats: these models have read the literature of every domain we test, so recall and derivation are not fully separable. The protein cohort is curated candidates rather than a random draw, held-out counts are small and nothing is bootstrapped. An effort setting is not a measure of compute. Frozen protocols, programs and research logs are kept for every run.